PROJECT FARO | DESIGN PROPOSAL

> > > Project Faro

Ask your terminal data. Understand the answer.



> > > TL;DR

The client connects the data their staff may use. Maria selects her account and asks a question in plain English. She does not need to write a glossary, rename columns or tell Faro how to join files.

Faro uses connectors: a connector is the part that knows how to read a particular source, such as a database, a report or a spreadsheet. Faro can connect to a client's existing data lake, but the client does not need to build one. It explains what it found, where it came from and what the records cannot establish.

In the demo, one client provides three files. The database records idle minutes, a spreadsheet names the berth for each posting, and a PDF explains which postings count. Faro uses all three to answer Maria. If a posting has no berth, it explains why it cannot name a winner.

Why Faro?

Faro means lighthouse. Its job is to show the operator what the records support and where the evidence stops.

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> > > 1. The client connects data. Maria asks.

Maria asks: **“Which berths had the most crane idle hours last month?”** She knows the operational question. She should not need to know whether the answer lives on page six of a report or across two tables.

What Faro prepares behind the scenes

The client grants access to the data and asks questions. The client does not prepare a glossary or a list of allowed question types. Faro stores only the connection settings and access permissions needed to reach the sources safely. It discovers their contents itself.

Faro lists only the files Maria may see. It can inspect their headings, read their records and point back to the original page, sheet or database result. These three examples show why finding a file and understanding its contents are different steps.

What Faro sees	How it uses the information
An October operations report	Index its headings and table locations. A table labelled “idle hours by berth” is a promising place to look; its month, units, coverage and version still need checking.
Two equipment names	A spreadsheet says C07; a database says QC-7. Look for a shared equipment ID or an existing register that links them. Similar spelling alone is not enough to combine their hours.
Only a current berth assignment	That can locate the crane today. It cannot put last month’s hours on a berth. Keep the terminal records usable and leave the historical berth unknown.

When to ask Maria

Ask only when her preference could change the answer. “Most idle hours” is already clear. If she says “worst” and total hours and idle percentage give different winners, show the distinction or ask which she wants. Do not ask her to map the database.

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> > > 1.1 One product, interchangeable connectors

Each connector answers the same three requests. The client keeps its infrastructure; the connector translates Faro’s request into a read that its source supports. The rest of Faro does not need to know whether that means querying a database or opening a workbook.

What Faro asks	What the connector returns
List what Maria may see	Names and references for permitted sources only. Access is checked for this client and user before discovery or reading.
Describe what is there	Native field names, location and file version. Units and meanings come from the source material when present; a column heading alone does not prove them.
Read the needed part	Requested values with a precise reference: record IDs, PDF page, or workbook sheet and row. Include the source version or snapshot when available; say when freshness or coverage cannot be checked.

What changes when the next client connects

For a supported source, configure its location, credentials and access scope. The connector handles its file layout or query method. A previously unseen format may need a reader built once and reused for other clients. Names and evidence can differ even when the format is the same.

What stays in Faro

Gate checks access; Scout finds candidate sources; Bridge decides what to read; Shield checks whether those facts support the answer. None needs a special rule for Client A’s berths. Faro’s own collection of turns, checks and cache entries stays separate from the client’s source material.

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> > > 1.2 What runs today and what comes next

The local demo reads JSON, SQLite, Excel and text PDFs through a separate Connector service. Its readers open the files. They do not decide which entries count as idle time or which terminal performed better.

Service	Working in the demo	Still to build
Scout | find where to look	Indexes connected field names and PDF passages; returns matches and permitted sources.	Expand content extraction and indexing for large sources, with background refresh.
Bridge | get the facts	Reads source notes and records. Can link two tables by matching fields and compare the resulting values.	Support larger sources, scanned reports and relationships that depend on overlapping time intervals.
Shield | check the answer	Repeats the reads and checks the proposed findings against their source versions and rows.	Test against real client revisions, access policies and independently reviewed answers.

Current boundary

The local demo selects a saved user account; production sign-in is still needed. SQLite, Excel and PDF access is granted per file. JSON records can also be restricted by an access field configured in Connector. Reads stop above 1,000 rows. The expression worker is isolated on macOS; other operating systems need an equivalent worker.

What stays reusable across clients

Clients keep their own formats and meanings. Faro standardizes what accompanies an answer: the measure, scope, dates, source, version and known gaps. A new format may need a reader; it does not need a new conversation or checking service.

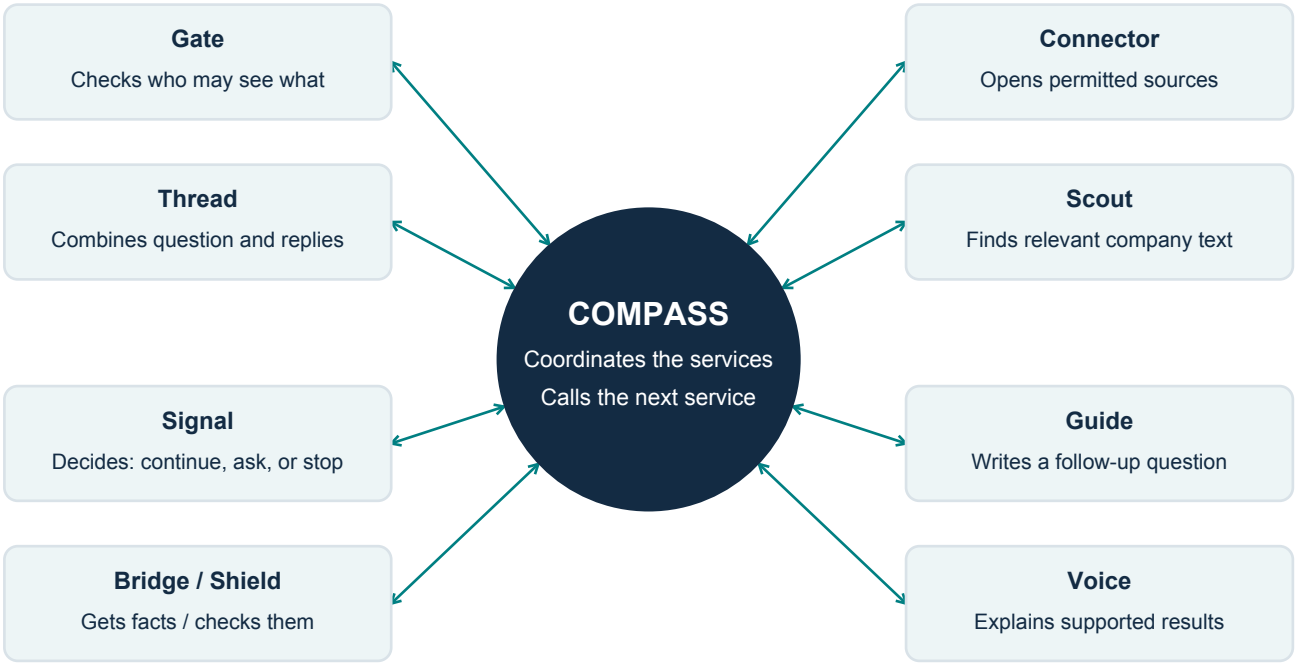
For changing records, follow the source’s revisions: count a correction once, exclude a record whose latest version marks it deleted, and recheck coverage when late entries arrive. If two versions cannot reliably be matched, do not present an exact total from them.

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> > > 2. From a question to an answer

Maria is an example terminal operator. She sees one screen. Compass coordinates the services below; it keeps her question, replies and evidence together.

Maria’s screen: sign in, ask, reply, read the answer



Connector: opens permitted files for Scout, Bridge and Shield.
Compass keeps the question, evidence and answer together in the company’s collection.

What comes in	What moves forward
Question and signed-in account	Gate supplies the person, company and permitted sites. Connector returns the sources those permissions allow.
Question and connected sources	Thread uses relevant replies; Scout searches source descriptions and passages. Signal decides whether to continue. Guide asks one question if a choice that matters is missing.
Clear request and permitted sources	Connector opens the files for Scout, Bridge and Shield. Bridge reads the data. Shield independently checks the evidence. Voice explains only supported findings. Compass saves the turn.

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> > > 2.1 Maria asks about crane idle time

Maria asks: **“Which berths had the most crane idle hours last month?”** The measure is clear. The open question is whether her records actually identify the berth. Being allowed to see two terminals does not establish either answer.

Step	Input → work → output
Gate | check access	Maria selects her demo account. Her account permits North and South at Harbor. Her words cannot grant access to another company. Connector lists only the connected sources her account may read.
Thread | understand replies	On the first turn, keep her question as asked. On later turns, combine her answer with the original request. Never borrow a choice from someone else’s conversation.
Scout | find relevant sources	Search field names and passages from the connected files. The October operations note is relevant: it explains the activity codes and links between the files. Return that passage with its source and version. No ranking has been produced yet.
Signal and Guide | clarify	Signal uses the Decision API only to classify the next step. If both total idle hours and idle share are legitimate interpretations, Guide asks which she means. If her wording or the source already settles it, proceed.

When a follow-up is useful

Maria already said “most crane idle hours,” so skip the follow-up. If she had said only “worst,” Guide could ask “Most total idle hours, or largest idle share?” Thread would keep her reply with the original question.

For a question asked in November 2026, “last month” means 1–31 October. Bridge interprets the question using the current reference date. The demo uses UTC; business timezone handling still needs to be connected for production.

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> > > 2.2 Read what Maria is allowed to see

Bridge receives the clarified question and the full catalogue of sources Maria may use. Scout's search results point to relevant source material; they do not restrict Bridge to one document or require every question to be a ranking.

One client, three sources

Source	What it adds to the answer
SQLite database	The current entries give the activity code and minutes for each work reference. The source provides a view that accounts for corrections and withdrawals.
Excel workbook	Each work reference is assigned to a berth. This lets Faro place the recorded idle time at the right berth.
PDF operations note	Page 1 identifies October, explains which code counts as idle, and says which references connect the two files. It contains no ranking.

Why all three matter

A database row gives minutes but no berth. The workbook gives a berth but no minutes. The note tells Faro when it is valid to combine them. Maria supplies none of these instructions herself.

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> > > 2.3 Follow one question across three files

1. Find the records

Scout returns matching source passages and the list of permitted sources. Bridge reads the October note. It learns that database field “ticket” and workbook field “Work reference” identify the same posting.

2. Connect the records

Bridge reads the current entries and the assignment sheet. It checks that the references match and that each posting has one berth. It groups the counted minutes by berth, then converts minutes to hours. Each read keeps the original location and file version.

3. Check the result

Shield opens the sources again and checks the evidence. With the complete October release, the ranking is S1: 115 hours, N1: 100, N2: 86 and S2: 71.

Change just one thing

Remove one assignment from the workbook and ask the same question. The database still has the idle minutes, but Faro cannot tell which berth should receive them. The console withholds the ranking. Maria is told why instead of being shown a different winner.

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> > > 2.4 Check the result before speaking

Shield receives the proposed findings and the records Bridge used. It repeats each read independently on the permitted source. It checks the source version, period, completeness, grouping and numbers; then it checks whether the words say more than the evidence.

What might go wrong	What the check does
The two terminals tie	Keep both at 186 idle hours. A request for one “worst” terminal is not evidence that one exists.
A record arrived late or was withdrawn	Re-read the current revisions. If a missing day could change the order, withhold the ranking. State which site or date is affected rather than calling the whole catalogue empty.
Rounding changes a decision	In October, South C01 made 4,678 moves in 234 operating hours: 19.99145 moves/hour. A screen rounded to 20.0 hides that it fell below a target of 20. Compare before rounding.

Voice | explain only what passed

“For 1–31 October, berth S1 recorded the most crane idle time: 115 hours. N1 followed at 100, N2 at 86 and S2 at 71.” The answer should make the indicator, dates, scope, source and limits visible. Compass retains the checked evidence for inspection.

Open the evidence

Below the answer, Faro returns the document name and a link. Connected files are saved as they were read, with the version retained. For shared JSON data, it returns only the permitted records, not another company’s records. Links open local files in the demo.

If part of a claim fails

Release an independently supported narrower finding if possible. If no finding can be verified, withhold the result and say why. A confident model answer is not a substitute for this check.

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> > > 2.5 A different question: why did dwell time spike?

Maria asks: **“Why did dwell time spike for reefer containers in week 12?”** A reefer is a refrigerated container. This question asks Faro to establish both a change and its cause. Those require different evidence.

Question to settle	What would support an answer
What is being compared?	Use start and end events documented in the source for dwell time, the client’s local week, the site and a comparison period. If the year or baseline matters and cannot be inferred, ask once.
Was there a spike?	Identify reefer containers, check the relevant event times and coverage in both periods, and compare using a measure supported by the source. A longer stay for one container does not prove a general spike.
Why did it happen?	To attribute a cause, require supporting event records tied to the affected containers and dates. A delay that happened at the same time is not proof that it caused the increase.

What Faro can say with the current Harbor catalogue

“These records cover crane activity, not reefer arrival and departure events. I can’t establish whether dwell time rose in week 12 or why.” Do not substitute crane downtime for container dwell time.

With a client that supplies dwell events but no reliable cause records, Faro can report the measured change and explicitly leave the cause unanswered. Faro explains that limit without asking Maria to reconcile files.

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> > > 3. Keep repeated questions affordable

Prepare the source guide once and refresh only what changes. Prefer a current reviewed report when it already answers the question; reading one table can avoid repeated work across raw files. Keep the same permission and evidence checks whichever route is used.

Where the saving comes from	What remains protected
Prepare once; search many times	Index the connected source descriptions and PDF passages. Reuse the saved index while source versions and access stay the same. In this demo the first search builds it; later searches embed only the question. The product should prepare it when data is connected.
Reuse checked data	While Maria is away, Luis asks the same question. Compass may use a saved read if the company, permitted scope, question, period and versions still match. Shield checks it again. A correction or changed access invalidates reuse.
Reuse model work	Keep successful model results in a separate company cache. Match the actual inputs, settings and permissions, including user scope. A clarification changes the inputs. A cache hit avoids the call; a failed result is not saved as a success.
Use only the model work needed	Signal uses a classification call. The shared model settings give different work different reasoning effort. Counts, sums and permission checks do not require a model to guess a number.

How to judge the trade-off

Track elapsed time, calls, retries, cache hits and cost per verified answer, by company. Review slow requests too, not just averages. Compare faster or cheaper settings against the same reviewed questions before adopting them.

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> > > 3.1 When a service times out

A slow model call does not justify restarting Maria’s question. Retry only the failed call, up to three attempts total, for a temporary provider failure. Do not repeat an access denial or a failed evidence check. If Shield cannot verify after retries, it returns no figures.

One saved demo turn

Maria requested a performance report for cranes by terminal. Signal classified the request once. Bridge made two successful model calls: two separate pieces of work, not two retries. Shield timed out on its first call, succeeded on the retry, then made a second successful call.

Service | job	Attempts in that turn
Signal | classify	1: success.
Bridge | read the facts	2: success; success.
Shield | verify findings	3: timeout → success on the same call; success on a separate call.

What the record tells us

The report was verified in about 95 seconds. The retry worked, but the operator still waited. This single demo run shows where time was spent; it does not establish typical speed, client savings or an acceptable wait.

Compass saves the question and turns. Each model service records its own attempts so a timeout remains visible even if its response never reaches Compass. Existing successful work stays attached to the original turn.

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> > > 4. Find out whether operators would use Faro

Choose recurring tasks with the terminal supervisor: preparing the morning report, comparing idle time or checking a target. Start with tasks people already complete by searching files and emailing colleagues. Success is reaching a sound operational decision with less effort.

Pilot step	Evidence to collect
Before the pilot	Ask operators to complete representative tasks the current way. Record elapsed time, handoffs and whether they found a usable answer. Have a data owner independently establish the expected numbers, scope and acceptable reasons to withhold.
Run both methods	Use comparable tasks and source versions. Include incomplete periods, ties, corrections, ambiguous names, changed permissions and unsupported causes. Review errors and unnecessary refusals, not only successful answers.
Watch the decision	Can the operator explain what the figures mean and what they do not prove? Would they act, ask for more data, or investigate? Record when an answer changes a decision and whether the reviewer agrees.
Decide whether to expand	Compare task completion, time saved, corrections, cost per completed task and repeat use without prompting. Agree thresholds with the client before the pilot. Stop rollout for unauthorized disclosure or materially misleading claims.

One concrete acceptance case

Use the same question twice. With complete assignments the correct winner is S1. With an assignment missing, the correct outcome is to explain the gap without naming a winner. Always test both halves: refusing everything is not useful either.

A repeatable demo proves the services can work together. Only a client pilot can show whether operators trust the answer enough to use it and whether it improves their work.

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> > > 4.1 Check the same design in other sectors

A reusable design must work when the client's vocabulary changes. The same services answered questions about terminal idle time, retail sales, hospital occupancy and hotel counts in the current console test. This is evidence of those examples, not proof of every possible source or sector.

Example	What the test established
Terminal operations	One question used a database, a workbook and a PDF, returning the supported berth ranking and the files used.
Retail	The model answered a total-sales question without a ranking requirement.
Hospitals	Six hospitals have July–October records covering occupancy, visits, waiting time, admissions, discharges and cancellations. Source gaps remain visible; Shield must not treat missing reports as zero.
Hotels	An inventory count worked without requiring a date range.

Stress the boundaries

Automated checks also use unfamiliar column names and a new recycling example. They test company isolation, duplicate or missing links, changed records, oversized reads and attempts to access files or the network. These are bounded checks, not a production security audit.

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> > > 4.2 Keep each decision understandable

The service names describe their work: Compass coordinates, Thread rewrites, Signal classifies, Guide generates a follow-up, Bridge reads, Shield verifies and Voice presents the answer. The shared model utilities keep their original prediction interface.

What the model returns

Each model prompt follows the same structure: objective, supplied information, initial analysis, final reasoning and response. The initial analysis briefly identifies relevant facts or gaps. The reasoning briefly explains the selected result. These summaries support review; they are not proof that an answer is correct.

What is saved

Each model-call record stores the service, turn, call ID, attempt outcomes and the assessment and reasoning returned for that attempt. Separate Bridge reads remain separate calls. A reused answer points back to its producing turn; the cache does not invent a new explanation.

Classification stays classification

Signal asks for the action, assessment and justification as three choices in the same Decision API call. It does not replace classification with a full free-text generation call. Other model responses use the Pydantic schema beside their prompt.

DATA COLLECTION | SQL DESIGN AND FIELD ORIGINS

> > > A.1 One row per conversation

Faro owns these SQL tables. Client files stay separate. One conversation has many turns; each saved result points back to the turn that produced it.

SQL column	Type	Where the value comes from
conversation_id	TEXT	Compass generates a UUID on the first message; later requests return that same ID.
company_id	TEXT	Gate: company on the signed-in demo account.
user_id	TEXT	Gate: verified account that started the conversation.
original_question	TEXT	First user message in supplied history, otherwise the first request.question.
created_at	TIMESTAMP	Compass UTC clock when the conversation starts.
access_json	JSON	Gate result at creation: user_id, company_id, permitted_scope_ids and permission_version from users.json.
initial_history_json	JSON	Only role and content supplied by the caller when starting a conversation; not independently verified history.

Maria’s example

Compass assigns the conversation ID. Gate supplies her company and account. Her clarification reply stays in that same conversation.

DATA COLLECTION | SQL DESIGN AND FIELD ORIGINS

> > > A.2 One row per turn

The turns table preserves what happened to each user message, including a clarification or failure.

SQL column	Type	Where the value comes from
turn_id	TEXT	Compass UUID for this individual message.
conversation_id	TEXT	Foreign key to the owned conversation selected by Compass after Gate checks access.
turn_number	INTEGER	Compass: previous saved turn count plus one, under the conversation lock.
message	TEXT	Exact current user message from request.question.
language	TEXT	Optional reply-language preference from the request, not an inferred language.
created_at	TIMESTAMP	Compass UTC clock before processing this turn.
status	TEXT	Compass: started or failed; otherwise Shield's status or the final next-action decision.
decision_source	TEXT	Compass identifies Signal or Bridge as the producer of the final action.
plan_json	JSON	Bridge's proposed source read, when available; null for a clarification before reading.
response_json	JSON	Service outputs collected by Compass; individual fields are listed in the field guide.
cache_events_json	JSON	Compass lookup events for this turn: service, collection, key, fresh/cache/rejected and producing turn.
model_attempts_tracked	INTEGER	Compass: tracking enabled for this turn. Older turns have no counts.
error	TEXT	Service, endpoint and failure type when a service call fails; otherwise the exception class. No invented substitute response.
elapsed_ms	INTEGER	Compass monotonic timer: milliseconds from turn processing start to outcome. Not an invoice.
cache_version	TEXT	Compass fingerprint of the deployed cache policy, service code and settings.

DATA COLLECTION | SQL DESIGN AND FIELD ORIGINS

> > > A.3 Two caches, one lookup pattern

ai_cache holds successful model results. data_cache holds evidence that passed Shield. A cache entry points back to its originating turn.

SQL column	Type	Where the value comes from
cache_key	TEXT	Compass hash of company, permissions, company information, policy version, service and exact inputs; AI keys also include the user.
company_id	TEXT	Gate: the company permitted to reuse this entry.
service	TEXT	Compass identifies the producing service call.
version	TEXT	Compass cache policy fingerprint when the entry was written.
cached_value_json	JSON	Typed output of the named AI service, or Bridge facts independently verified by Shield.
expires_at	TIMESTAMP	Compass UTC save time plus 120 days.
produced_by_turn	TEXT	Foreign key to the turn that actually produced this value; absent if an older record did not save it.

Keep the checks

Gate checks access again; Shield rechecks reused evidence. A change of company, permissions, wording, source contents, source version or model settings invalidates the relevant saved work.

DATA COLLECTION | SQL DESIGN AND FIELD ORIGINS

> > > A.4 Trace an answer and its model calls

Faro's SQL design stores one model_calls row per logical model call. The turn ID links it back to the original question. Two reads are separate calls; a retry is another attempt inside the same call.

SQL column	Type	Where the value comes from
turn_id	TEXT	Compass turn ID, passed to the service with the request.
call_id	TEXT	Shared utils generates one ID per logical model call; retries keep it.
service	TEXT	Name of the service making the model call.
started_ns	INTEGER	Service clock at the start of the logical call, Unix nanoseconds.
explanations_json	JSON	Model response: brief initial_analysis and reasoning per attempt, with its attempt number. Not private chain of thought.
outcomes_json	JSON	One outcome per provider attempt, in order; in_progress until the provider returns.

Field origins

SQL column	Type	Where the value comes from
table_name	TEXT	Declared Faro table whose field is described.
field_path	TEXT	Column name, or nested path in its saved JSON value; [] denotes an item.
source	TEXT	Description defined in the Faro schema or recorded service output contract.

Thread supplies the rewritten question. Scout supplies matching source passages and references. Signal classifies the next action; Guide asks the follow-up if needed. Bridge supplies reads and evidence. Shield supplies verified findings or a reason for withholding. Voice supplies the answer. Compass records the turn outcome and cache use.

How to read the attempt history

A timeout followed by success inside one call means the retry worked. No entry means the service made no model call on a tracked turn. Older turns without tracking are unknown, not zero.